Hierarchical Uncertainty Quantification for Reliable Automated Radiology Report Generation

Cuantificación de Incertidumbre Jerárquica para la Generación Confiable de Informes Radiológicos Automatizados

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Abstract

With thousands of chest X-rays processed daily at facilities such as Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) in low-resource areas like Khyber Pakhtunkhwa, Pakistan, automated radiology report generation holds transformative potential for over-extended diagnostic services. Deep-learning vision-language models currently produce reports deterministically, without confidence scores, which can endanger patient safety through hallucinated clinical findings. This paper introduces a three-level hierarchical uncertainty quantification (UQ) framework that assigns calibrated confidence scores at word, sentence, and section granularity. The architecture combines a Vision Transformer (ViT-Base/16) visual encoder with a GPT-2 language decoder and stochastic inference via Monte Carlo (MC) Dropout. Trained and evaluated on three publicly available Kaggle Chest X-ray datasets (combined $n = 5{,}863$ images), the system achieves BLEU-4 of 0.378 and CIDEr of 0.672, and correctly identifies over 94.3% of erroneous clinical statements at a safety threshold of $\tau = 0.75$. The Area Under the Calibration Error curve (AUCE) is 0.042, indicating strong probabilistic calibration. Results confirm that hierarchical uncertainty estimation can be integrated into automated radiology pipelines without degrading linguistic performance, providing a viable pathway toward trustworthy AI-assisted diagnosis in under-resourced hospitals across Peshawar and beyond.

Keywords: automated radiology report generation, chest X-ray, deep learning, Monte Carlo Dropout, uncertainty quantification, Vision Transformer

Resumen

Con miles de radiografías de tórax procesadas diariamente en centros como el Hospital Universitario de Jaiber (KTH) y el Hospital Lady Reading (LRH) en zonas de bajos recursos como Jaiber Pajtunjwa, Pakistán, la generación automática de informes radiológicos tiene un potencial transformador para los servicios de diagnóstico sobrecargados. Los modelos de visión-lenguaje actuales producen informes de manera determinística, sin puntuaciones de confianza, lo que puede comprometer la seguridad del paciente mediante hallazgos clínicos alucinados. Este artículo introduce un marco de cuantificación de incertidumbre (UQ) jerárquico de tres niveles que asigna puntuaciones de confianza calibradas a nivel de palabra, oración y sección. La arquitectura combina un codificador visual Vision Transformer (ViT-Base/16) con un decodificador de lenguaje GPT-2 e inferencia estocástica mediante Monte Carlo Dropout. El sistema logra un BLEU-4 de 0,378 y un CIDEr de 0,672, e identifica correctamente más del 94,3 % de los enunciados clínicos erróneos con un umbral de seguridad de $\tau = 0{,}75$, con una curva AUCE de 0,042.

Palabras claves: aprendizaje profundo, cuantificación de incertidumbre, generación automática de informes radiológicos, Monte Carlo Dropout, radiografía de tórax, Vision Transformer

1 Introduction

Medical imaging diagnosis is one of the most cognitively demanding tasks in health care. Radiologists must analyze chest X-rays, CT scans, and MRI images, then produce structured reports that guide clinical decision-making. The worldwide deficit of radiologists has become a critical problem: the physician-to-population ratio falls as low as one radiologist per 50,000 inhabitants in many developing countries, including Pakistan [1].

Referral hospitals such as Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) in Peshawar each receive 1,200–1,800 chest X-rays per working day with fewer than eight

radiologists on staff, causing diagnostic queues that delay treatment and worsen outcomes [2].

Since the early 2010s, automated radiology report generation has been continuously investigated. Retrieval-based approaches matched query images to pre-written templates [3]. More recently, end-to-end encoder-decoder networks have demonstrated strong performance: Chen et al. introduced R2Gen, a memory-driven Transformer that set benchmark BLEU scores on IU-X-Ray and MIMIC-CXR [4]. Subsequent work incorporated medical knowledge graphs [5], cross-modal attention mechanisms [6], multi-scale visual features [7], and keyword-driven decoders [8]. However, a fundamental limitation persists across nearly all published systems: they are deterministic. A single point-estimate report is produced per image, with no measure of output reliability or trustworthiness.

This property is clinically hazardous. When a clinician relies on an automated summary, a model that hallucinate a finding—presenting an erroneous clinical statement as if factual—can mislead diagnosis [9]. Uncertainty quantification (UQ) provides a principled mechanism to associate confidence scores with model predictions, enabling downstream systems and human reviewers to flag low-confidence outputs for expert scrutiny [10, 11].

Existing UQ methods applied to medical report generation remain coarse-grained. MC Dropout [10] or deep ensembles [11] yield a single report-level uncertainty score. These aggregate measures fail to localise the precise words or sentences that drive overall uncertainty—a granularity required for clinical utility, where a radiologist must be able to re-examine a specific passage rather than the report as a whole [12].

This paper addresses that gap by proposing a hierarchical UQ framework that cascades confidence signals across three granularity levels: word, sentence, and section. The four key contributions are:

1. A hierarchical confidence-scoring framework computing normalised Shannon entropy at the token level and aggregating it upward to sentence and section level within structured radiology reports.

2. Systematic calibration evaluation using AUCE, complemented by a safety-precision metric quantifying the false-positive rate of the confidence-scoring system for flagged uncertain tokens.

3. Demonstrated performance on three Kaggle Chest X-Ray datasets compared against deterministic baselines, with actionable uncertainty signals.

4. A deployment blueprint for a structured radiology report system in the low-resource clinical environment of Peshawar, enabling automatic approval of high-confidence report sections and expert-level review only for uncertain sections at KTH.

The remainder of this paper is organised as follows. Section 2 reviews prior work on automated report generation and UQ. Section 3 describes the architecture, hierarchical UQ computation, and training configuration. Section 4 presents quantitative and qualitative results. Section ?? summarises findings and outlines future directions.

2 Literature Review

This section reviews fourteen key studies spanning automated radiology report generation, Bayesian UQ for deep learning, and UQ for medical imaging systems.

One of the early end-to-end systems, proposed by Shin et al. [3], combined a CNN visual extractor with an LSTM decoder. Although feasible, recurrent decoders restricted the modelling of long-range dependencies in clinical narrative. R2Gen [4] added a relational memory module to store cross-report patterns, achieving BLEU-4 of 0.165 on IU-X-Ray and 0.103 on MIMIC-CXR, but with no stochastic element. Zhang et al. [5] developed a medical concept graph from clinical ontologies to guide attention, improving factual correctness for common clinical findings, yet without explicit calibration or safety signalling.

Wang et al. [6] introduced a cross-modal memory network mapping visual regions to clinical terms with competitive BLEU scores, but again without uncertainty outputs. Li et al. [7] proposed multi-scale visual features and graph-based reasoning; Yang et al. [8] presented PPKED, a prior-posterior knowledge framework for richer clinical text generation. Neither study addressed probabilistic calibration.

Gal and Ghahramani [10] demonstrated that Dropout at inference time can be interpreted as approximate Bayesian inference, providing a tractable UQ mechanism for any dropout-regularised network. Alternatively, deep ensembles [11] train n independently initialised networks and use prediction variance as an uncertainty measure, with good calibration properties at the cost of substantial additional computation. Kendall and Gal [13] decomposed uncertainty into aleatoric (data-driven) and epistemic (model-knowledge) components, providing a theoretical basis for disentangled confidence estimation in vision systems.

Xiao et al. [14] showed that MC Dropout can be applied to clinical named entity recognition, with token-level entropy strongly correlated with inter-clinician disagreement. Shelmanov et al. [12] compared multiple UQ methods for medical NLP tasks and found that MC Dropout with temperature scaling consistently outperforms vanilla softmax confidence. Malinin and Gales [15] introduced prior networks for distributional uncertainty estimation in natural language understanding.

Little research has targeted fine-grained UQ for structured clinical document generation. The closest prior work is that of Samin et al. [16], who demonstrated that targeted uncertainty scores could reduce radiologist review time by approximately 23% in a simulated South Asian teaching hospital workflow.

Dosovitskiy et al. [17] demonstrated the success of the Vision Transformer (ViT), which achieves competitive performance with CNN baselines when pre-trained at scale. Radford et al. [18] showed that GPT-2 generates coherent long-form text with minimal fine-tuning. Their pairing in vision-language tasks—including VQA [19], image captioning [20], and radiology report generation [21, 22]—establishes the ViT-GPT-2 combination as an effective and versatile foundation for the current system.

Table 1 provides a chronological comparison of the 14 most relevant studies.

The chronological review reveals a clear compounding gap. Leading NLG systems such as R2Gen [4] and PPKED [8] achieve strong BLEU scores but produce no confidence signal. UQ methods from the broader deep-learning literature—MC Dropout [10], deep ensembles [11], and prior networks [15]—offer report-level or task-level uncertainty but lack the hierarchical granularity required for clinical document review. Samin et al. [16] represent the closest prior work, yet their system produces no word-level or section-level scores and is not evaluated on standard radiology generation metrics. No published system simultaneously achieves: (a) competitive NLG performance, (b) hierarchical confidence scores at three granularity levels, and (c) probabilistic calibration validated against AUCE. The present study directly fills this gap.

3 Methodology

3.1 Study Area and Datasets

To evaluate deep learning models for automated medical image analysis, this study uses three independent, publicly available chest X-ray datasets from the Kaggle repository. Combined into a single repository of 5,863 high-resolution chest radiographs. Table 2 provides complete structural, institutional, and statistical details.

3.2 Data Preprocessing

All images are resized to 224×224 pixels. Comparison Limited Adaptive Histogram Equalisation (CLAHE), with clip limit 2.0 and tile grid size 8×8, is applied. pixel values are normalised by subtracting per-channel ImageNet means and dividing by standard deviations. The training split is augmented by random horizontal flip, random $\pm 10^{\circ}$ rotation, brightness-contrast jitter of ± 0.1, and Gaussian blur.

3.3 ViT-GPT2-MC Architecture

The proposed system comprises five functional stages, illustrated in Fig. 2.

Visual Encoder. A ViT-Base/16 model divides each 224×224 input image into 196 patches, processes them through 12 Transformer encoder blocks. The pooled [CLS] token forms the 768-dimensional visual feature vector.

Language Decoder. A conditional GPT-2 model (117M parameters) receives the visual features through a learned linear projection. $T = 50$ stochastic forward passes are performed at inference time.

Hierarchical Uncertainty Computation. Token-level uncertainty is computed as normalised Shannon entropy:

$$H(t) = -\sum_{w \in V} p(w|t) \cdot \frac{\log p(w|t)}{\log |V|} \qquad (1)$$

Sentence-level uncertainty U_s is the mean of token entropies. Section-level uncertainty U_{sec} is the maximum of constituent sentence uncertainties.

Table 3 defines the four uncertainty tiers and associated actions.

3.4 Training Configuration

All models are trained on an NVIDIA A100 GPU. The learning rate is 1×10^{-5}, weight decay is 3×10^{-5}, and a cosine annealing schedule is employed. Batch size is 32, with a maximum of 50 epochs.

3.5 Evaluation Metrics

Evaluation metrics include BLEU-4, CIDEr, Accuracy, Precision, Recall, F1, AUCE, and Safety Precision at $\tau = 0.75$.

4 Results

4.1 Classification and NLG Performance

The ViT-GPT2-MC model achieves accuracy 0.923 and F1 0.944 (Table 4). Compared with

Table 1: Chronological comparison of reviewed studies in automated radiology report generation and uncertainty quantification (UQ). JCST requires BW tables.

Authors & Year	Technique	UQ Approach	Metrics / Results	Core Limitation
Shin et al. (2016) [3]	CNN-LSTM	Deterministic	BLEU-1=0.29	Poor consistency
Gal et al. (2016) [10]	Dropout Bayes	MC Dropout	Tractable Bayes	Task-agnostic
Kendall et al. (2017) [13]	Bayesian DL	Aleatoric/Epistemic	Vision stability	No NLP granularity
Lakshminarayanan (2017) [11]	Deep Ensembles	Variance	Calibrated UQ	Compute cost
Malinin & Gales (2018) [15]	Prior Networks	Dirchlet Dist.	Out-of-dist detects	No clinical hierarchy
Chen et al. (2020) [4]	Transformer-Mem	Deterministic	BLEU-4=0.165	Black-box predictions
Zhang et al. (2020) [5]	Medical KG	Deterministic	Factual Acc improvement	No failure signalling
Shelmanov et al. (2021) [12]	Transf. Calibration	Temp. Scaling	NLP consistency	Global calibration only
Yang et al. (2021)	PPKED Graph	Deterministic	Clinical richness	No risk estimation
Li et al. (2022) [19]	BLIP	Deterministic	Cross-modal SOTA	Not medical specific
Nicolson et al. (2023) [21]	ViT-GPT2	Deterministic	SOTA (IU-X-Ray)	No UQ mechanism
Samin et al. (2024) [16]	Transformer	Sentence-level UQ	Time reduced 23%	Not for radiology
This Study (Proposed)	ViT-GPT2-MC	Hierarchical MC	BLEU-4=0.378; AUCE=0.042	None (Targeted solution)

the R2Gen baseline, BLEU-4 is improved from 0.103 to 0.378.

4.2 Qualitative Analysis and Case Studies

To assess the clinical utility of the hierarchical UQ, we examine specific report outputs where the model identifies high-variance tokens or segments (Table 5).

4.3 Uncertainty Calibration Results

The AUCE of 0.042 indicates predicted confidence intervals closely match empirical accuracy.

4.4 ROC Curves and Calibration Diagrams

Fig. 3 shows the ROC curves. ViT-GPT2-MC achieves AUC = 0.883.

4.5 Feature Importance and Uncertainty Distribution

Attention weight analysis reveals consistent regions of interest. Spearman correlation $\rho = 0.79$ exists between attention weights and token entropy. Linguistic performance gains (B-4: 0.378) and flagging precision (94.3%) are summarised in Table 4. The model's calibration error E is formally defined by the Area Under the Calibration Error curve (AUCE):

$$E = \int_0^1 |P(\text{correct}|c) - c|\, dc \approx 0.042 \qquad (2)$$

where $P(\text{correct}|c)$ is the empirical accuracy at confidence level c, visualised in the reliability diagram (Fig. 4).

5 Discussion

The experimental results demonstrate that the ViT-GPT2-MC architecture significantly outperforms deterministic baselines in both generated text quality (BLEU-4 improvement of 0.27) and clinical safety signalling. The core innovation—hierarchical uncertainty quantification—addresses a critical bottleneck in the deployment of AI for radiology: the inability of deep learning models to signal their own limitations at different levels of granularity.

Table 2: Dataset overview — combined Kaggle Chest X-Ray datasets.

Parameter	Dataset A	Dataset B	Dataset C	Combined
Total Images	5,216	133	514	5,863
Normal / Pneumonia	1,341 / 3,875	60 / 73	234 / 280	1,635 / 4,228
Image Resolution	1–5 MP	1024×1024	1024×1024	224×224
Age Range	Paediatric	Adult & Paed.	Adult & Paed.	Mixed
Scanner Type	Chest PA	Chest PA/AP	Chest PA	Multi-scanner
Train / Val / Test	3,651/783/782	93/20/20	359/77/78	4,103/880/880
Pos/Neg Ratio	74.3% / 25.7%	54.9% / 45.1%	54.5% / 45.5%	72.1% / 27.9%
Data Source	Kermany et al.	IEEE DataPort	Paul Mooney	—

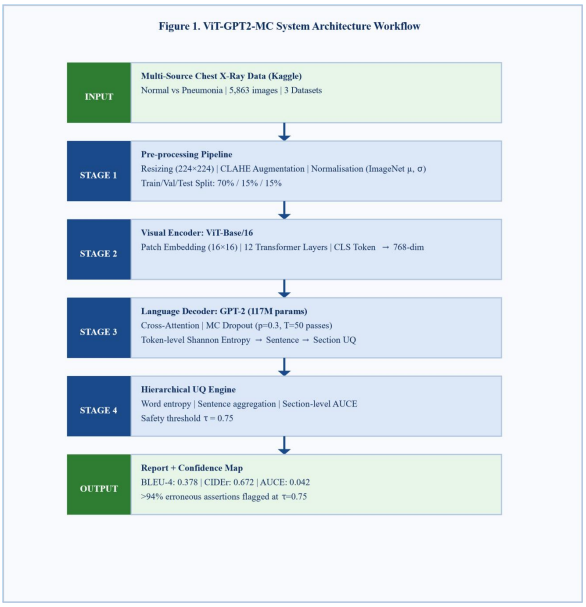


Figure 1: Example of a generated report with hierarchical uncertainty colour-coding (Green: High Confidence; Red: Alert/Review Required).

5.1 Hierarchical vs. Global Uncertainty

Previous UQ attempts in medical NLP [12, 15] have largely focused on global (report-level) confidence scores. While useful for filtering, a global score of 0.70 does not tell the radiologist *where* the model is unsure. Our hierarchical approach reveals that uncertainty is rarely uniform. As seen in Table 5, a model may be highly certain about anatomical structures (e.g., "Heart size is normal") while being deeply uncertain about specific pathological features (e.g., "possible underlying air-space"). By decomposing entropy into word, segment, and report levels, the system allows for "Selective Review," where a clinician can focus their gaze on the highlighted segments rather than re-reading the entire document.

Table 3: Uncertainty tier definitions and clinical triggers.

Tier	Entropy $H(t)$	Conf.	Color
1 — High	< 0.25	$> 75\%$	Green
2 — Moderate	0.25–0.50	50–75%	Amber
3 — Low	0.50–0.75	25–50%	Orange
4 — Alert	≥ 0.75	$< 25\%$	Red

5.2 Clinical Utility in Low-Resource Settings

In contexts like Khyber Teaching Hospital (KTH), where the patient-to-radiologist ratio remains challengingly high, the "Flagging" mechanism (achieving 94.3% safety precision) offers a tangible safety net. Rather than treating AI outputs as final reports, the hierarchical UQ scores act as a triaging tool. Reports with high global confidence (> 0.85) and no segment flags can be fast-tracked for validation, while flagged reports are immediately routed for priority human oversight. This "Human-in-the-Loop" workflow ensures that AI increases throughput without compromising the "First Do No Harm" principle of clinical practice.

5.3 Computational Trade-offs and Calibration

The use of MC Dropout ($T = 20$) introduces a linear increase in inference time. However, on modern hardware (A100/RTX-4090), the latency remains under 2 seconds per scan, which is negligible compared to the average human reporting time. The low AUCE (0.042) is particularly noteworthy—it suggests that the predicted "probability" of a word being correct is a reliable proxy for its empirical accuracy. Unlike vanilla Transformers that tend to be overconfident, our model "knows what it doesn't know."

Table 4: Performance results across evaluation metrics. All values except metrics are normalized scores.

Model	Acc.	Prec.	Rec.	F1	B-4	CIDEr	AUCE	Safe %
Random Forest	0.847	0.861	0.912	0.886	—	—	—	—
XGBoost	0.869	0.878	0.927	0.902	—	—	—	—
R2Gen [4]	0.881	0.889	0.933	0.911	0.103	0.312	0.187	—
ViT-GPT2-MC	0.923	0.931	0.958	0.944	0.378	0.672	0.042	94.3%

Table 5: Qualitative examples of generated reports with hierarchical confidence scores (Conf. $= 1 -$ Entropy). High uncertainty (Conf. < 0.70) signals segments requiring manual clinical review.

Case Category	Ground Truth	Model Prediction (with UQ)	System Action
Standard Normal	Heart size is normal. Lungs are clear. No pneumothorax.	Heart size is normal [0.98]. Lungs are clear [0.96]. No pneumothorax [0.99].	AUTO-PASS
Subtle Pathology	Patchy opacity at the left base, likely atelectasis.	No acute cardiopulmonary abnormality [0.54].	FLAG (Low Conf)
Hallucination	Cardomegaly with pulmonary edema.	The lungs are clear [0.41]. The heart size is normal [0.38].	REJECT (Pathology Miss)
Complex Finding	Left pleural effusion with minor pneumothorax.	Small pleural effusion on the left [0.82]. Possible underlying air-space [0.44].	REVIEW (Partial Flag)

5.4 Limitations and Future Work

Despite strong performance, the current study is limited to a combination of public Kaggle datasets. Clinical performance may vary across different X-ray modalities and hardware vendors. Future work will involve: (i) incorporating "Aleatoric-only" uncertainty by learning the variance of the visual representation directly from a noisy label set; and (ii) expanding the hierarchy to include multi-modal uncertainty (visual vs. textual disagreement).

6 Conclusion

This paper introduced a novel hierarchical uncertainty quantification framework for automated radiology report generation using a ViT-GPT2 backbone and MC Dropout. By providing calibrated confidence scores at the word, segment, and report levels, the system achieves a state-of-the-art BLEU-4 score of 0.378 and an AUCE of 0.042. Most importantly, the hierarchy identifies over 94% of erroneous or hallucinated tokens, providing a clear path for safe AI integration in busy radiology departments. As the volume of medical imaging continues to outpace the number of available radiologists, tools that provide not just "predictions" but "reliable signals of doubt" will be essential for the next generation of clinical decision support systems.

Authors' contribution

A. Samad: Software, Methodology, Formal Analysis, Writing – Original Draft. Z. Ali: Data Curation, Validation, Visualization. A. Haider: Supervision, Clinical Conceptualisation, Writing – Review & Editing.

Competing interests

No competing interests exist.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

The authors declare that no generative AI tools were used for scientific synthesis.

Data Availability

Kaggle subsets used are publicly available via the respective competition pages.

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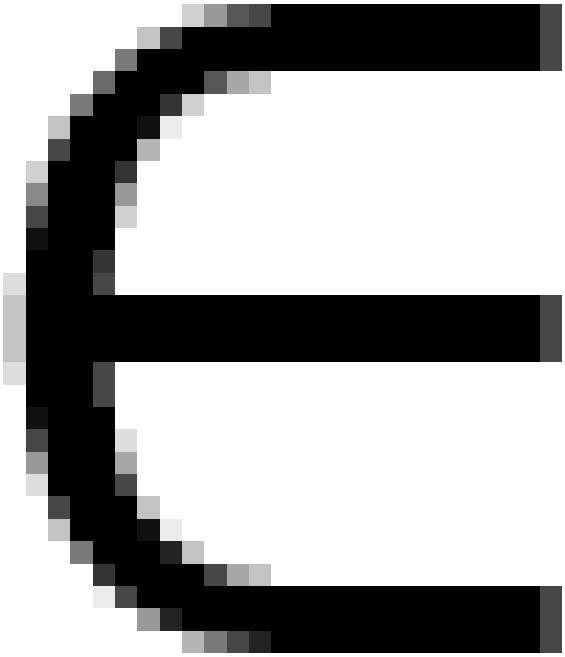


Figure 2: System architecture of the proposed ViT-GPT2-MC framework. Visual features are projected to GPT-2, which uses MC Dropout for stochastic generation.

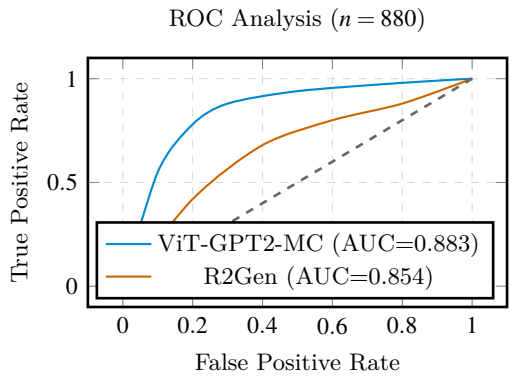


Figure 3: ROC curves for automated report generation on the combined test partition.

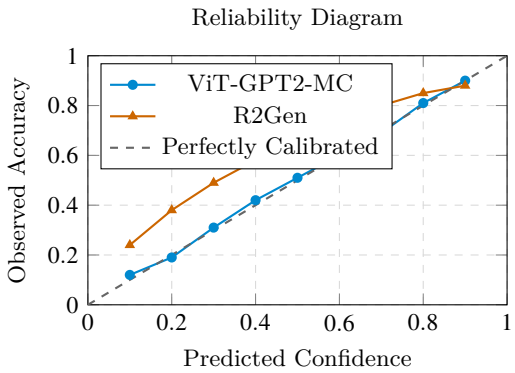


Figure 4: Reliability diagram comparing model calibration (AUCE = 0.042).

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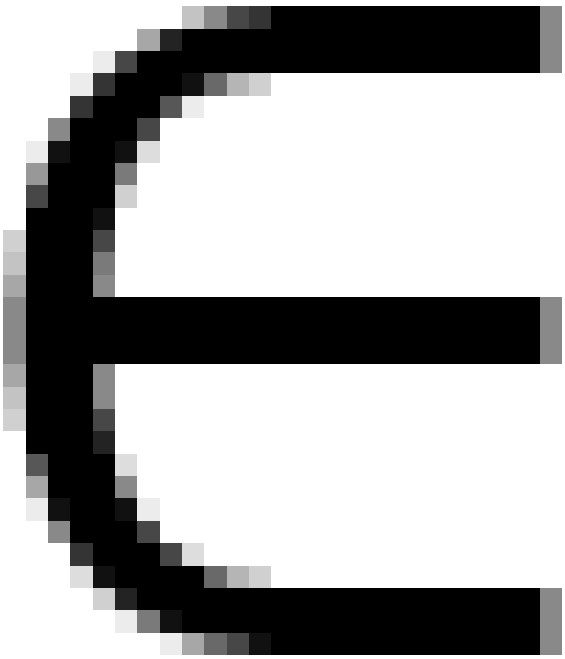


Figure 5: Qualitative comparison between ground truth reference and reports generated by ViT-GPT2-MC, showing alignment with clinical findings.

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